Appendix: Artifact Description

Artifact Description (AD)

Paper ID: `pap142`

I. Download

```
git clone https://github.com/TempGNN-Program/
  ↪ TempGNN.git
cd TempGNN
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

II. Environment

```
Board: AMD/Xilinx Alveo U280,
  ↪ xcu280-fsvh2892-2L-e
Platform: xilinx_u280_gen3x16_xdma_1_202211_1
OS: Ubuntu 22.04 x86_64
XRT: 2.16.204
Vitis/Vivado: 2023.2 (only for rebuilding
  ↪ xclbins)
GCC: 11.4.0
GNU Make: 4.3
Python: 3.10 or newer
```

III. Run on U280

```
source /opt/xilinx/xrt/setup.sh
make ae-core-u280 U280_CORE_DEVICE=0
  ↪ U280_CORE_REPETITIONS=3
```

The command runs TempGNN, MATG, ViTeGNN, and RTGA. The expected arithmetic mean latency over six datasets and four models is:

U280 implementation	Mean latency (ms)
TempGNN	1.296553
MATG	9.966618
ViTeGNN	2.869752
RTGA	4.192824

Fresh measurements are written under `results/reviewer_u280_runs/<run-id>/`.

IV. Generate Figures from result.csv

All remaining figure values are read from `results/result.csv`.

```
python3 -m scripts.reproduce_paper_figures
```

The generated CSV/SVG files are written to `results/paper_reproduction/`.

V. Baseline Status

- MATG: independently reproduced from the IPDPS 2022 paper and its partial public HLS headers; includes a distinct source, runner, and U280 xclbin.
- ViTeGNN: independently reproduced from the TPDS 2025 paper; includes a distinct source, runner, and U280 xclbin.
- RTGA: independently reproduced from the DAC 2024 paper; includes a distinct source, runner, and U280 xclbin.
- Cascade and TGLite-CPU: retained only as packaged paper comparison records; no fresh execution of those external stacks is claimed.
- TempGNN: includes HLS kernels, testbenches, Vitis build scripts, XRT host code, U280 board logs, and its own U280 xclbin.

The three baselines are independent, paper-based forward-path reproductions, not the authors' complete original stacks.

Run `make release-preflight` before creating the Zenodo archive. The separate `make release-preflight-results` target additionally requires a paper-equivalent configuration and numerical tolerance PASS. See `ZENODO_RELEASE_CHECKLIST.md`.

VI. Metric Meaning

`DDTC` is dependency-driven TDP construction. `OATS` is overlap-aware TDP synchronization.

`packet_reuse_factor` measures how many per-target packet materializations collapse into unique PHLE packets. `memory_bytes` is packet state/metadata traffic. `cycles` is the kernel's cycle proxy used by C-sim and exported fixture stats. HLS reports provide C/RTL latency after synthesis/cosim.

VII. License

This artifact is licensed under the Apache License 2.0. See `LICENSE`.